



Unit Test - Form B

Introduction to Matter

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

Directions: Select the best answer for each question. Mark one choice only. Complete this test after all lessons and the cumulative review.

1. Which observation is not evidence that something is matter?

- A. It occupies volume.
- B. It has mass.
- C. It can be weighed.
- D. It is visible.

2. A model contains only identical particles made of two different atom types. What is the best classification?

- A. Compound
- B. Element
- C. Homogeneous mixture
- D. Heterogeneous mixture

3. What does a particle model most directly help students inspect?

- A. The exact particle color
- B. The actual particle size
- C. Composition and distribution
- D. The sample's odor

4. A clear liquid leaves two recoverable substances after a physical separation. It is best classified as a

- A. pure element
- B. homogeneous mixture
- C. pure compound
- D. single particle

5. Which sample is a compound?

- A. Distilled H₂O
- B. Clean copper
- C. Well-mixed air
- D. Granite



Unit Test - Form B

Introduction to Matter

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

6. Which sample is a homogeneous mixture?

- A. Clean copper wire
- B. Distilled water
- C. Granite
- D. Well-mixed air

7. Which sample is heterogeneous?

- A. A single pure crystal
- B. Oil and water layers
- C. Distilled water
- D. Clean aluminum

8. Which result would most strongly challenge a claim that a sample is pure?

- A. It has a constant temperature.
- B. It looks uniform.
- C. A physical method separates two components.
- D. It has a measurable volume.

9. Which pair contains two physical properties?

- A. Density and conductivity
- B. Flammability and reactivity
- C. Rusting and burning
- D. Reactivity and color

10. Which property is extensive?

- A. Density
- B. Melting point
- C. Volume
- D. Color



Name: _____ Date: _____ Period: _____

11. Why is flammability a property even before ignition?

- A. It describes potential chemical behavior.
- B. A flame is always invisible.
- C. Properties require no evidence.
- D. Burning is a physical state.

12. Which process is a chemical change?

- A. Salt dissolves and is recovered.
- B. A tablet is crushed.
- C. Water freezes.
- D. Wax burns and forms new substances.

13. Which observation alone is insufficient to prove a chemical change?

- A. A new substance is identified.
- B. Bubbles appear.
- C. Product composition differs.
- D. Atoms are rearranged into products.

14. Why is crushing a solid usually a physical change?

- A. Its size changes without requiring a new substance.
- B. Its mass becomes zero.
- C. Its atoms disappear.
- D. Its chemical formula must change.

15. Which method separates two immiscible liquid layers?

- A. Chromatography
- B. Magnetism
- C. Decantation or a separatory funnel
- D. Filtration through paper only



Name: _____ Date: _____ Period: _____

16. Why does filtration not remove dissolved salt from water?

- A. Salt is magnetic.
- B. Water is a solid.
- C. The salt reacts with paper.
- D. Dissolved particles pass with the liquid.

17. Which property difference is central to distillation?

- A. Color
- B. Volatility or boiling behavior
- C. Magnetism
- D. Particle shape only

18. Which sequence is best for dry iron filings mixed with sand?

- A. Use a magnet, then evaluate the sand.
- B. Distill before any other step.
- C. Use chromatography first.
- D. Melt both components.

19. A uniform blue liquid leaves no filter residue but yields liquid and solid by distillation. What is the best claim?

- A. It is an element.
- B. It is a pure compound.
- C. It is a homogeneous mixture with dissolved material.
- D. Filtration destroyed the solute.

20. Which statement correctly applies conservation of matter?

- A. Only visible products count.
- B. Gas products count even if they leave an open container.
- C. Matter can disappear during heating.
- D. Conservation applies only to physical changes.